

SECURITY · AI · REAL-TIME

WatchService Agent

Real-time Ransomware Detection System

Behavior-based AI detection · Instant alert · Gemini AI guidance

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System Overview

What problem does it solve?

⚠️ The Problem

Ransomware encrypts files

and demands ransom — data loss is irreversible

Traditional AV misses

new/unknown variants
(signature-based limits)

No real-time alert

Users notice too late;
damage already done



✓ Our Solution

Behavior-based Detection

Monitors file activity patterns — not file content. Works on any ransomware variant.

XGBoost AI Model

9 behavioral features (read/write/delete/entropy change) → SAFE / WARNING / DANGER

Instant Multi-channel Alert

Fullscreen emergency popup + email notification the moment DANGER is detected

Gemini AI Guidance

LLM generates step-by-step response guide tailored to the detected threat

Key Features

Core capabilities of WatchService Agent

Real-time File Monitoring

- Java NIO WatchService
- Recursive folder watching
- CREATE / MODIFY / DELETE events
- 3-second window aggregation

AI-powered Detection

- XGBoost binary classifier
- 9 behavioral feature vectors
- Trained on 30,000 samples
- SAFE / WARNING / DANGER labels

Entropy Analysis

- Multi-section sampling (40/40/20%)
- Encryption detection via entropy spike
- Noise filter: $|diff| > 0.05$ threshold
- file_encrypt_like_count feature

Instant Alert System

- Fullscreen emergency popup
- Email notification (SMTP)
- Gemini LLM response guidance
- Dashboard with history & stats

Persistent Settings

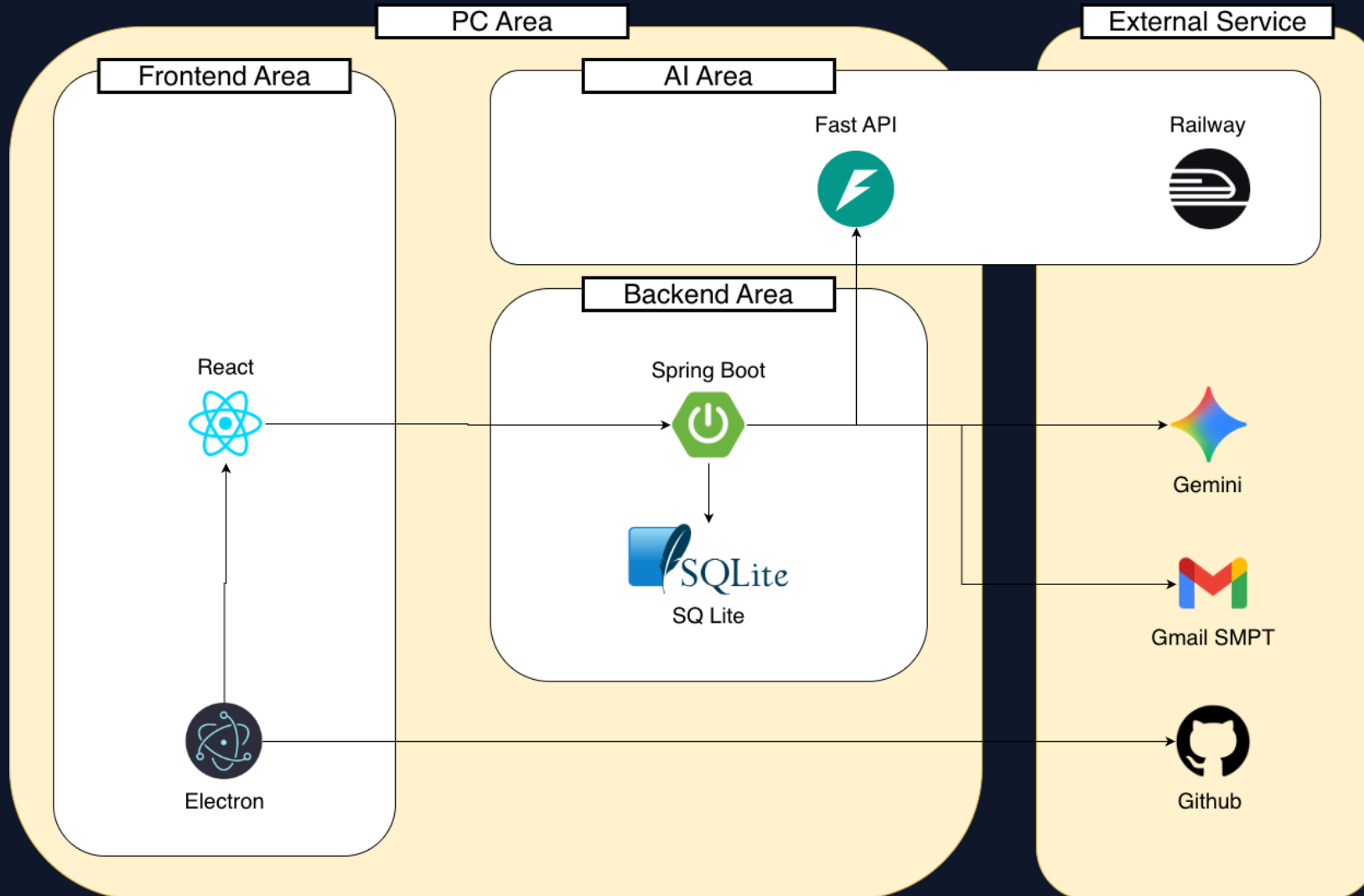
- Fixed ownerKey 'default'
- Survives server restart
- Watch folders + exception rules
- Alert email configuration

Reporting & Admin

- PDF report export
- CSV / JSON log download
- Admin dashboard
- False-positive reporting

System Architecture

System components and flow



Demo

Live demonstration scenario

Step 1

Launch & Setup

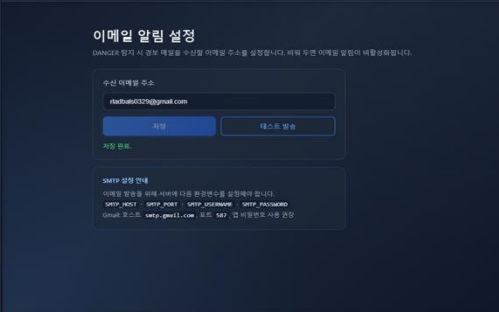
Open WatchService Agent → Start monitoring
Settings → Add watch folder (e.g. Desktop/test)



Step 2

Configure Email Alert

Settings → Alert Email → Enter recipient
Send test email → Verify delivery in inbox



Step 3

Simulate Normal Activity

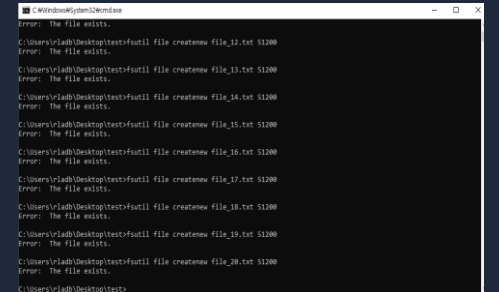
Create / edit / delete a few text files
Dashboard shows SAFE (green) — no alert



Step 4

Simulate Ransomware Behavior

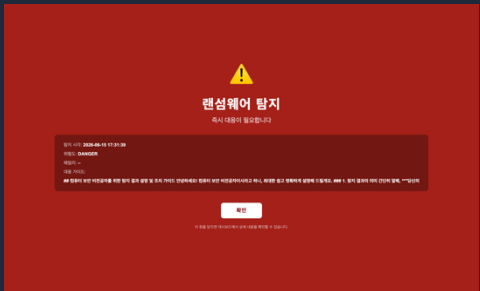
Run simulate script: rapid mass rename +
high-entropy overwrite on 20+ files



Step 5

DANGER Detection

Fullscreen emergency popup appears
Email alert delivered to inbox



Step 6

Response & Review

Gemini AI guidance displayed on screen
Dashboard → Notification detail → history log

